

Generation Interconnection System Impact Study Report

For Queue Project Q12-0190

Helm'S Pumped Storage 138 kV

August 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0190. The Interconnection Customer proposes a 1212.5 MW natural gas combined cycle facility interconnecting at the Helm'S Pumped Storage 138 kV substation in the Maryland area. The request was received on December 29, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 131 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$51,466,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Helm'S Pumped Storage / Glenannan 138kV, the Helm'S Pumped Storage / Mcewen 138kV loads to 512.5% of its 232 MVA rating. Project DFAX: 100.0%.

For the loss of Mcewen / Helm'S Pumped Storage 138kV, the Helm'S Pumped Storage Substation / Glenannan 138kV loads to 495.6% of its 240 MVA rating. Project DFAX: 100.0%.

Seat Pleasant / Mcewen 138kV: 273.7% loading (225 MVA) following the outage of the Helm'S Pumped Storage / Glenannan 138kV Line; DFAX = 60.0%.

Sunny Hills / Mcewen 138kV: 216.2% loading (249 MVA) following the outage of the Helm'S Pumped Storage / Glenannan 138kV; DFAX = 40.0%.

Bladensburg / Sunny Hills 138kV: 212.6% loading (240 MVA) following the outage of the Helm'S Pumped Storage / Glenannan 138kV; DFAX = 40.0%.

For the loss of Glenannan / Helm'S Pumped Storage 138kV, the Central Avenue / Mt Lassen 138kV loads to 144.3% of its 204 MVA rating. Project DFAX: 24.4%.

Mt Lassen / Seat Pleasant 138kV line: 142.1% loading (205 MVA) following the outage of the Helm'S Pumped Storage / Glenannan Substation 138kV line; DFAX = 24.4%.

The Beltsville / Bunker Hill 138kV is overloaded to 137.4% of its 229 MVA emergency rating for the loss of the Glenannan / Taber 345kV. The project contributes 15.4% DFAX (186.5 MW) to this overload.

The Seat Pleasant / Hartford 138kV L is overloaded to 135.3% of its 221 MVA emergency rating for the loss of the Helm'S Pumped Storage / Glenannan 138kV. The project contributes 35.6% DFAX (432.0 MW) to this overload.

For the loss of Bunker Hill / Nasa 138kV, the Siuc / Fairland Substation 138kV loads to 131.9% of its 181 MVA rating. Project DFAX: 5.8%.

For the loss of Glenannan / Nasa 138kV Cir. 2, the Glenannan / Nasa 138kV line loads to 129.7% of its 257 MVA rating. Project DFAX: 9.6%.

Glenannan / Nasa 138kV: 129.7% loading (257 MVA) following the outage of the Glenannan / Nasa 138kV Cir. 1; DFAX = 9.6%.

The Rush / Campus Drive 138kV is overloaded to 128.9% of its 158 MVA emergency rating for the loss of the Helm'S Pumped Storage / Glenannan 138kV. The project contributes 21.2% DFAX (256.8 MW) to this overload.

For the loss of Glenannan Substation / Helm'S Pumped Storage 138kV line, the Campus Drive / Bladensburg 138kV Line loads to 128.4% of its 225 MVA rating. Project DFAX: 21.2%.

Kingswood / Hartford 138kV L: 122.6% loading (227 MVA) following the outage of the Helm'S Pumped Storage / Glenannan 138kV line; DFAX = 35.6%.

The Bunker Hill / Nasa 138kV is overloaded to 121.6% of its 290 MVA emergency rating for the loss of the Taber / Glenannan 345kV Line. The project contributes 15.4% DFAX (186.5 MW) to this overload.

Taber 345/138kV Auto: 110.2% loading (551 MVA) following the outage of the Taber / Dixleta 345kV; DFAX = 7.3%.

Kingswood / Suitland 138kV: 108.3% loading (248 MVA) following the outage of the Helm'S Pumped Storage / Glenannan 138kV; DFAX = 18.3%.

For the loss of Taber / Glenannan 345kV, the Sierra Pacific Lumber 138 kV/1 kV Autotransformer loads to 107.6% of its 528 MVA rating. Project DFAX: 7.9%.

For the loss of Dixleta / Taber 345kV L, the Dixleta / Drum Substation 138kV loads to 107.4% of its 274 MVA rating. Project DFAX: 7.2%.

Summit Lake / Bladensburg 138kV: 105.9% loading (167 MVA) following the outage of the Glenannan / Helm'S Pumped Storage 138kV L; DFAX = 18.8%.

For the loss of Tombstone / Bixby Hill 500kV line, the SharyInd / Bixby Hill 345kV loads to 103.4% of its 1,414 MVA rating. Project DFAX: 8.4%.

The Lake Creek Substation / Sierra Pacific Lumber 138kV is overloaded to 100.6% of its 322 MVA emergency rating for the loss of the Taber / Glenannan 345kV. The project contributes 6.1% DFAX (73.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcewen / Helm'S Pumped Storage 138kV	Rebuild line with higher-capacity conductor	\$7,064,000	100% (\$7,064,000)
Helm'S Pumped Storage Substation / Glenannan 138kV	Rebuild line with higher-capacity conductor	\$3,293,000	100% (\$3,293,000)
Mcewen / Seat Pleasant Substation 138kV	Rebuild line with higher-capacity conductor	\$780,000	100% (\$780,000)
Sunny Hills / Mcwen Substation 138kV	Rebuild line with higher-capacity conductor	\$1,388,000	100% (\$1,388,000)
Sunny Hills / Bladensburg 138kV	Rebuild line with higher-capacity conductor	\$2,657,000	100% (\$2,657,000)
Central Avenue / Mt Lassen 138kV	Rebuild line with higher-capacity conductor	\$682,000	100% (\$682,000)
Mt Lassen / Seat Pleasant 138kV line	Rebuild line with higher-capacity conductor	\$352,000	100% (\$352,000)
Beltsville / Bunker Hill 138kV	Rebuild line with higher-capacity conductor	\$1,221,000	63% (\$767,000)
Hartford / Seat Pleasant 138kV	Rebuild line with higher-capacity conductor	\$3,213,000	100% (\$3,213,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Siuc / Fairland Substation 138kV	Rebuild line with higher-capacity conductor	\$1,599,000	100% (\$1,599,000)
Glenannan / Nasa 138kV line	Rebuild line with higher-capacity conductor	\$5,673,000	100% (\$5,673,000)
Gleannan / Nasa 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,667,000	100% (\$2,667,000)
Campus Drive Substation / Rush 138kV	Rebuild line with higher-capacity conductor	\$1,963,000	100% (\$1,963,000)
Campus Drive / Bladensburg 138kV Line	Rebuild line with higher-capacity conductor	\$5,049,000	100% (\$5,049,000)
Kingswood Substation / Hartford 138kV	Rebuild line with higher-capacity conductor	\$2,667,000	100% (\$2,667,000)
Nasa / Bunker Hill 138kV	Rebuild line with higher-capacity conductor	\$2,582,000	63% (\$1,623,000)
Taber 345/138kV Auto	Replace transformer with higher-rated unit	\$15,606,000	30% (\$4,726,000)
Kingswood / Suitland 138kV	Reconductor line	\$281,000	100% (\$281,000)
Sierra Pacific Lumber 138-1 kV Auto	Replace transformer with higher-rated unit	\$6,501,000	30% (\$1,937,000)
Drum Substation / Dixleta 138kV	Reconductor line	\$1,041,000	31% (\$320,000)
Summit Lake / Bladensburg 138kV	Reconductor line	\$376,000	100% (\$376,000)
SharyInd / Bixby Hill 345kV	Reconductor line	\$16,865,000	13% (\$2,118,000)
Lake Creek Substation / Sierra Pacific Lumber 138kV	Reconductor line	\$527,000	52% (\$271,000)
	Total allocated to project		\$51,466,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.